

Non-Minimum-Phase Channel DoA Estimation via UCA Phase-Mode Smoothing and Enhanced Subspace Separation

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Chinmay Deo¹, C. Ramanathan², C. R. Venkatesh², K. Giridhar²

National Institute of Technology Jalandhar, India¹

Indian Institute of Technology Madras, India²



- A “**non-minimum phase**” channel is when non-LOS signal components arrive at Rx with higher power.
- Multipath creates coherent sources, leading to **rank deficiency** of the data covariance matrix in **subspace based** DoA estimation methods.
- Traditional model order selection (AIC/MDL) [1] criteria fail in this scenario.

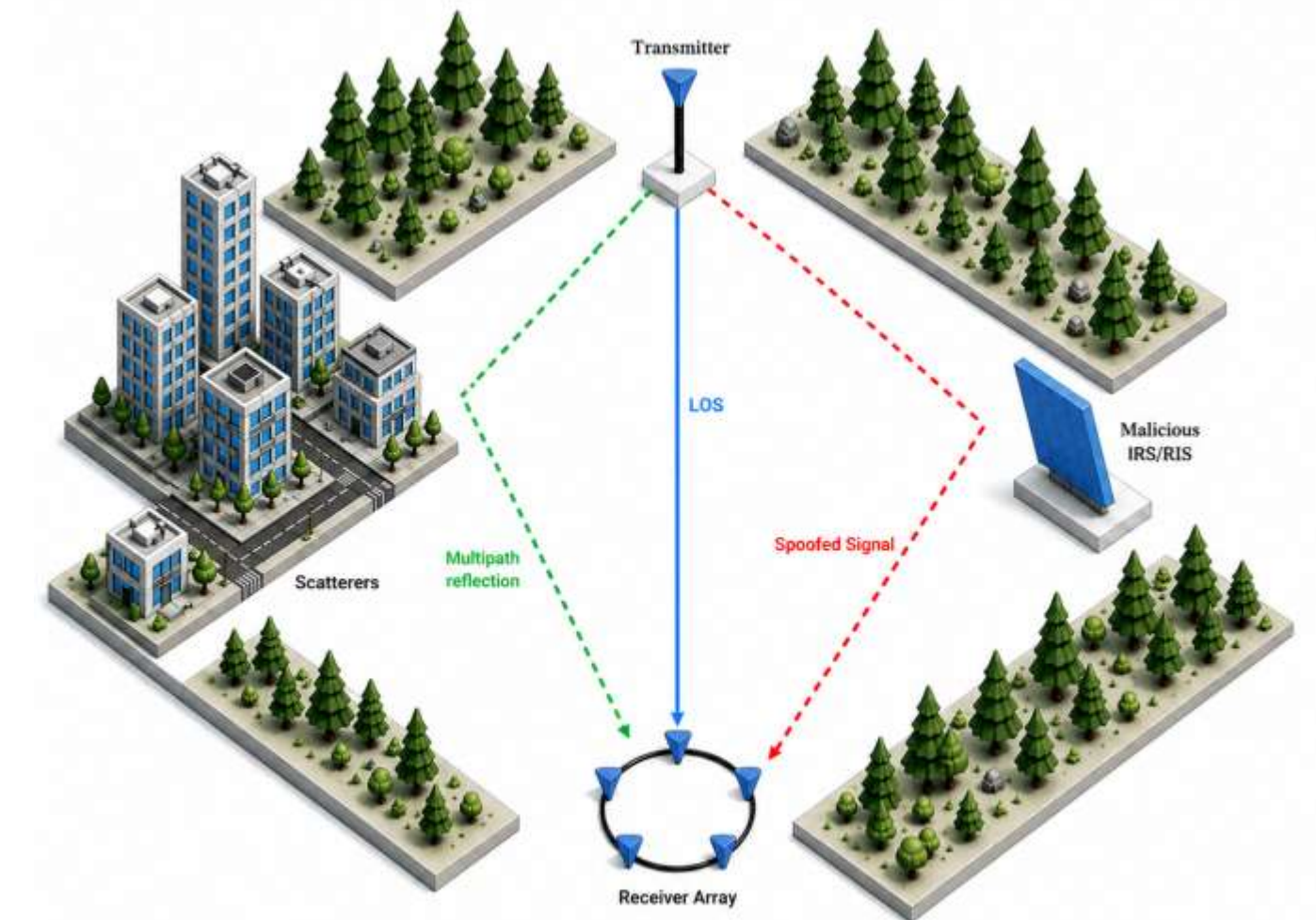


Fig. 1: Illustration of non-minimum phase scenario

[1] M. Wax and T. Kailath, “Detection of signals by information theoretic criteria,” IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 33, no. 2, pp. 387–392, 1985

- Beamformer weighting scheme [2]: $w(\gamma_n) = e^{jm\gamma_n}$, where $\gamma_n = \frac{2\pi(n-1)}{N}$

- Far-field array pattern for mode m ($N > 2M$) becomes
$$A_m(\phi) = j^m \cdot J_m(\zeta) \cdot e^{jm\phi}$$

Where 'N' is number of UCA elements, 'M' is maximum mode order, $J_m(\zeta)$ is the Bessel function of the first kind (order m), $\zeta = K_0 r \sin \theta$, K_0 is wave number, θ is elevation and ϕ is azimuth.

- After **beam-space transformation**, the UCA manifold is mapped onto an M' -dimensional **virtual ULA**. The resulting virtual ULA is partitioned into $S = M' - M'' + 1$ overlapping M'' -dimensional subarrays for mode-space **spatial smoothing** [3].

[2] C. Mathews and M. Zoltowski, "Eigenstructure techniques for 2-d angle estimation with uniform circular arrays," IEEE Transactions on Signal Processing, vol. 42, no. 9, pp. 2395–2407, 1994.

[3] R. Eiges and H. Griffiths, "Mode-space spatial spectral estimation for circular arrays," IEE Proceedings - Radar, Sonar and Navigation, vol. 141, pp. 300–306, 1994

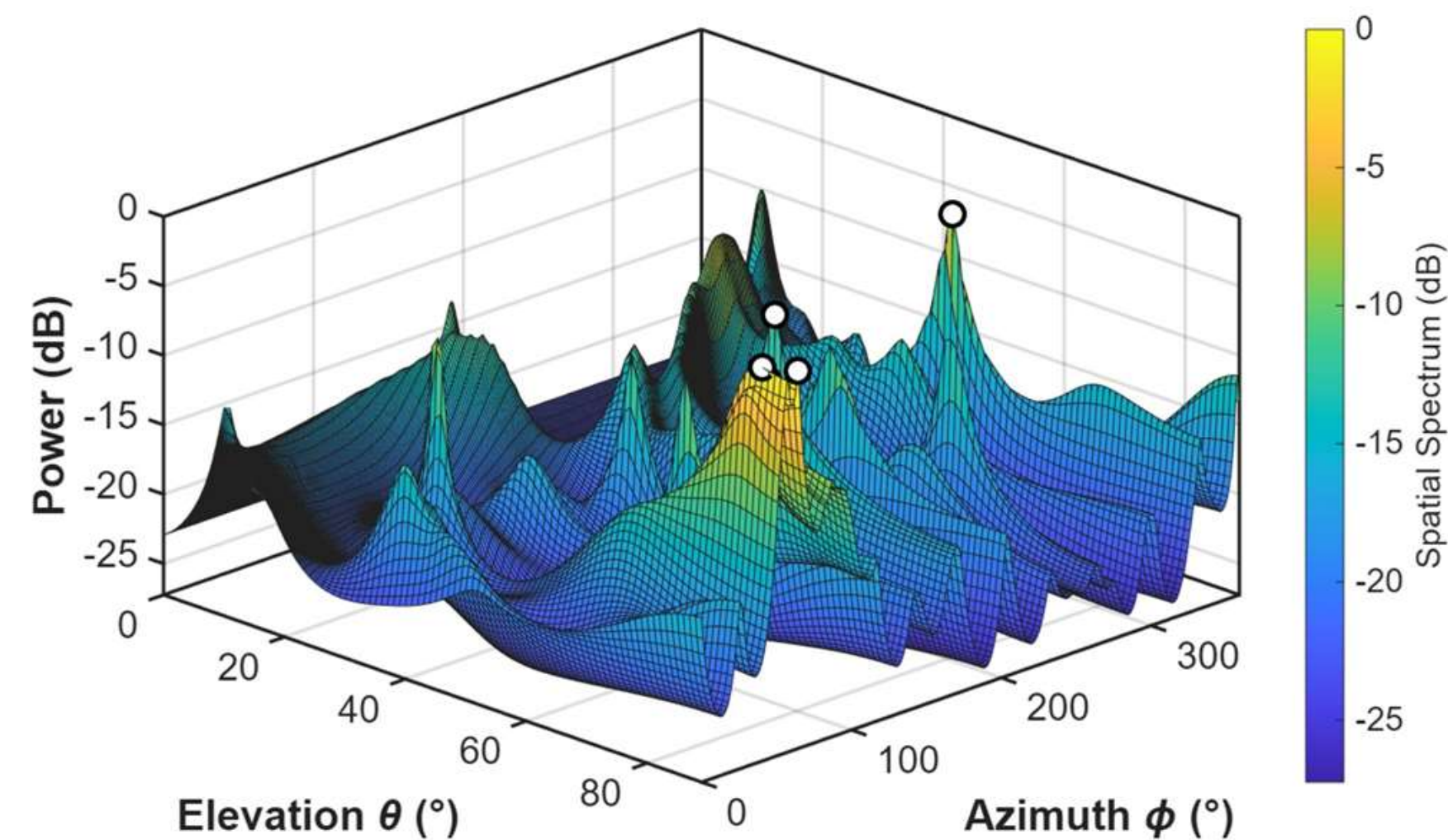


Fig. 2: Spatial Spectrum using AIC estimate

Ground Truth Signal Scenario:

Source	Φ (Az)	Θ (El)	Power	Delay
S1 (LOS)	55	90	-3 dB	0
S2	65	90	0 dB	1 sample
S3	85	90	-6 dB	2 samples
S4	100	90	-12 dB	3 samples

Parameters:

Carrier: $f_c = 702$, **Geometry:** $N = 5$ elements, $r = 0.5095m$, **SNR:** $9dB$,
Snapshots: $T = 128$, **No. of subarrays for smoothing:** $S = 2$

Case	Model Order Selection Criterion	Estimated peaks (φ , θ)	Power (dB)
Unsmoothed	Manual	(65, 82)	0
		(273, 29)	-2.12
		(333, 89)	-2.41
Smoothed	AIC	(65, 90)	0
		(57, 85)	-0.15
		(101, 75)	-4.19
	MDL	(65, 90)	0
		(55, 89)	-0.08
		(101, 89)	-4.58
	Our Method	(55, 90)	0
		(62, 89)	-0.71
		(269, 26)	-1.98

Table I: Comparison of Dominant Peak Estimates Under Different Model-Order Selection Methods

- Appropriate choice of S (Subarray aperture) for mode-space smoothing and K (Noise variance multiplier) enables estimation of lowest delay path DoA.

- $$\underbrace{(M' - S + 1) \left\{ \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} \right\}^{-1}}_{\text{Theoretical estimate of smallest signal eigenvalue}} \geq (K - 1) \sigma_n^2$$

Theoretical estimate of smallest signal eigenvalue

- $C_{ij} = \begin{cases} 1, & i = j \\ \rho_f(\theta_i, \theta_j), & i \neq j \end{cases}$, where $|\rho_f| = \left| \rho \frac{\sin(S(\theta_i - \theta_j)/2) \cos \psi}{S \sin((\theta_i - \theta_j)/2)} \right|$ is post-smoothing source correlation matrix and smoothed signal covariance matrix is $\overline{R}_s = P^{1/2} C P^{1/2}$ with source powers $P = \text{diag}(P_1, \dots, P_D)$ for D number of coherent sources.

- The noise variance is estimated from the smallest p eigenvalues:

$$\hat{\sigma}_n^2 = \frac{1}{p} \sum_{i=M''-p+1}^{M''} \lambda_i$$

- The dynamic signal subspace boundary is then defined as

$$\gamma = K \cdot \hat{\sigma}_n^2$$

- Eigenvalues satisfying $\lambda_i > \gamma$ are classified as belonging to the restored signal subspace.

- Monte Carlo simulations using spatially white Gaussian noise were used to estimate the empirical cumulative distribution function (CDF) of the normalized noise eigenvalues $\lambda_i / \hat{\sigma}_n^2$, where $\hat{\sigma}_n^2$ denotes the tail-eigenvalue, which is taken as a conservative noise variance estimate.

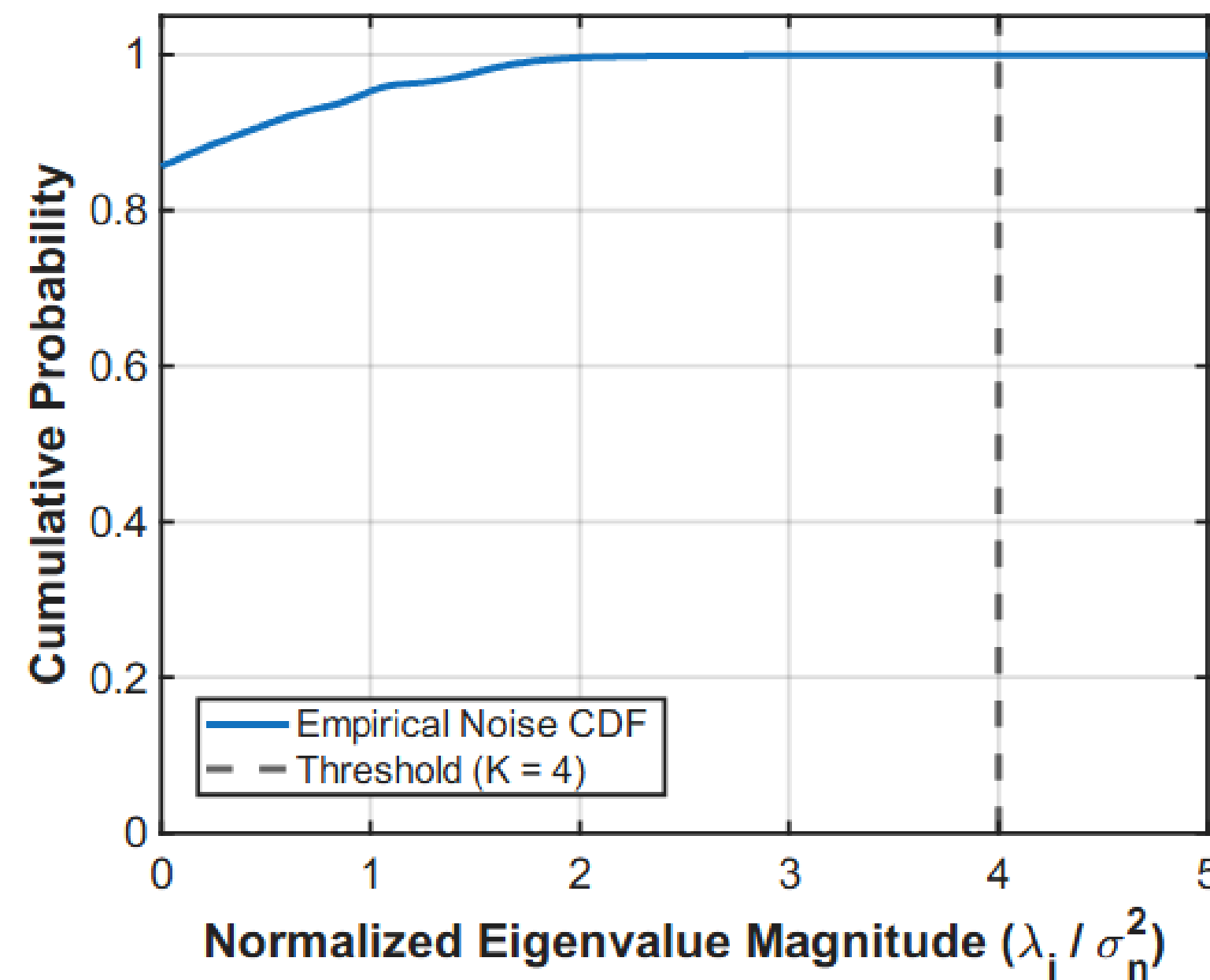
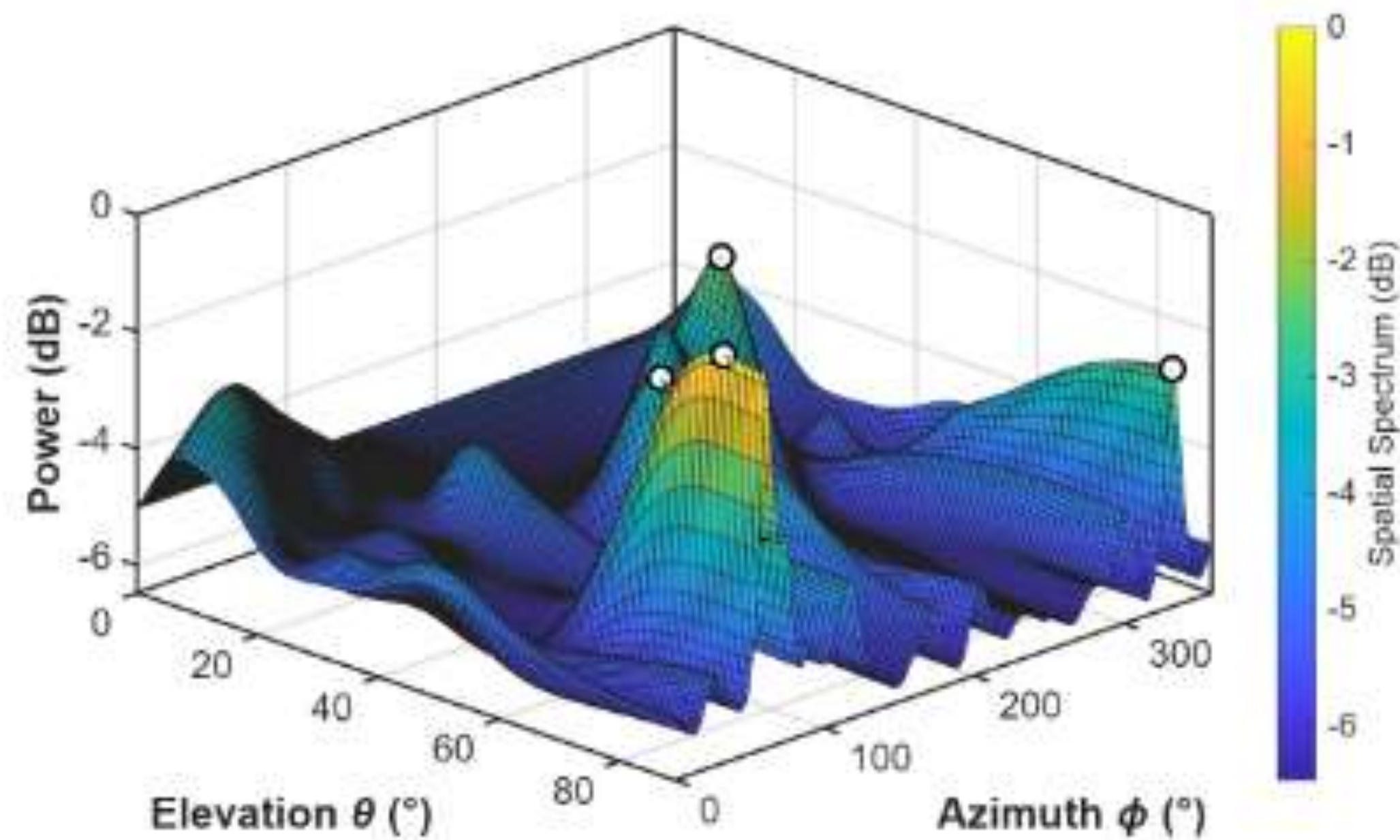


Fig. 3: Empirical CDF of normalized noise eigenvalues after phase-mode smoothing and forward-backward averaging

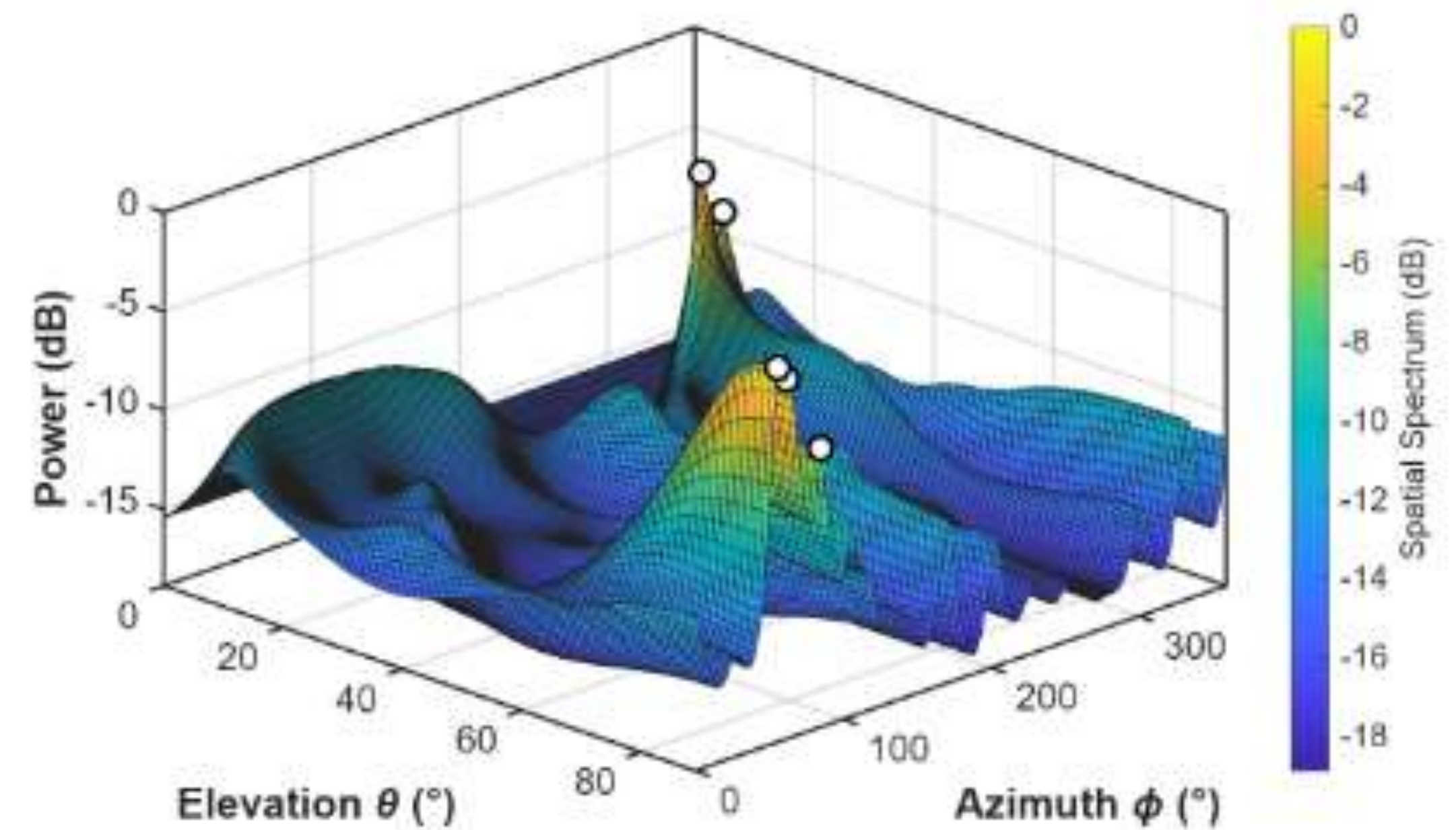
RESULTS

Smoothed vs Un-smoothed Pseudospectra

- As summarized in table I and Fig. 4, the smoothed variant with dynamic thresholding successfully isolates LOS DoA as the dominant peak.



(a) MUSIC spectrum without spatial smoothing.



(b) MUSIC spectrum after spatial smoothing.

Fig. 4: Comparison of MUSIC pseudo-spectra in the presence of coherent multipath.

Smoothed vs Un-smoothed Eigenvalue Distributions

- After smoothing, four dominant eigenvalues became clearly separable enabling proper subspace partitioning using the proposed thresholding scheme (Fig. 5).

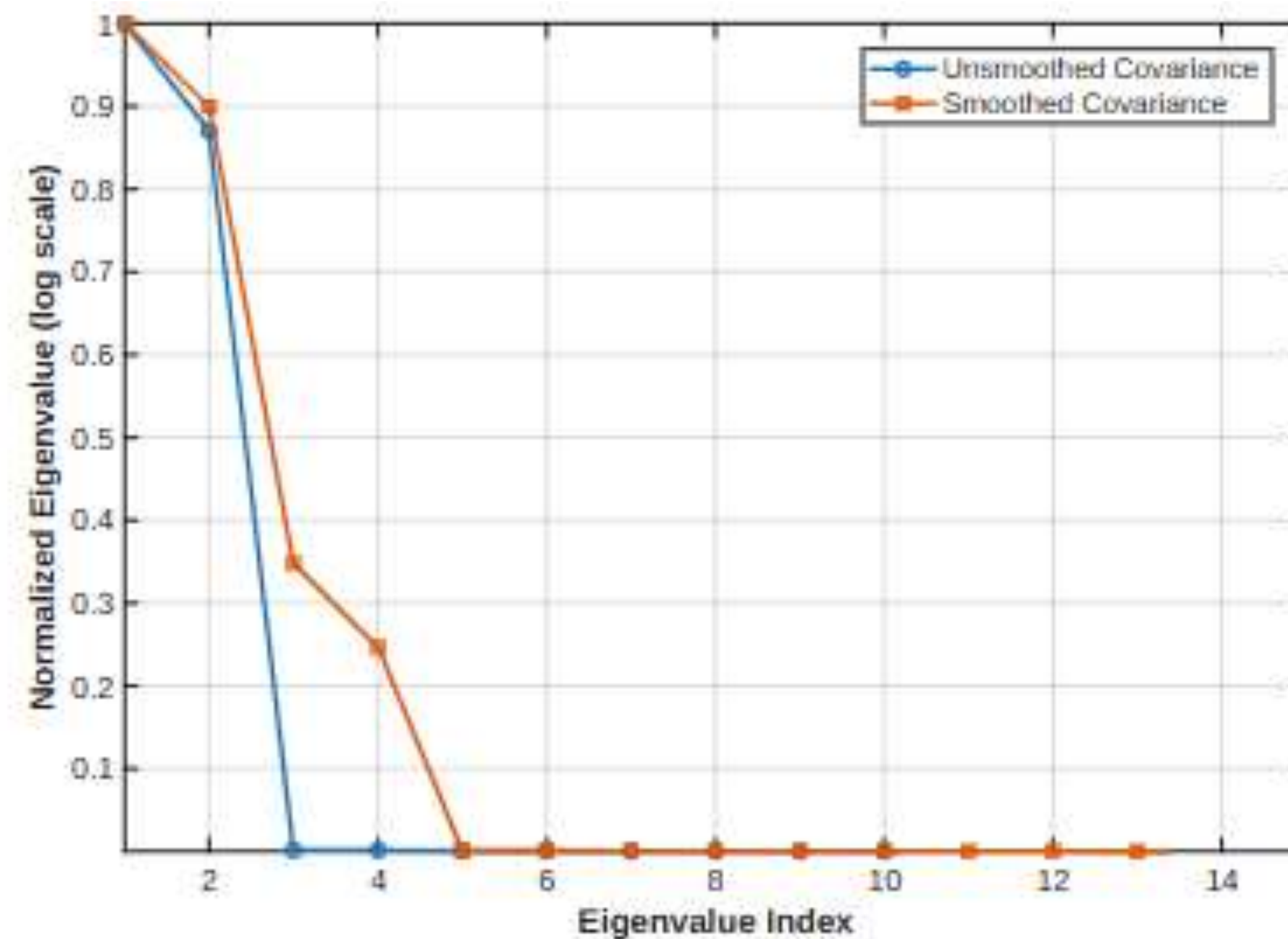


Fig. 5: Eigenvalue distribution before and after spatial smoothing

- Fig. 6 shows the performance of the algorithm over a range of SNRs. Each point represents an average of 400 Monte-Carlo trials. Beyond 5 dB operational SNR, our method approaches the CRB.

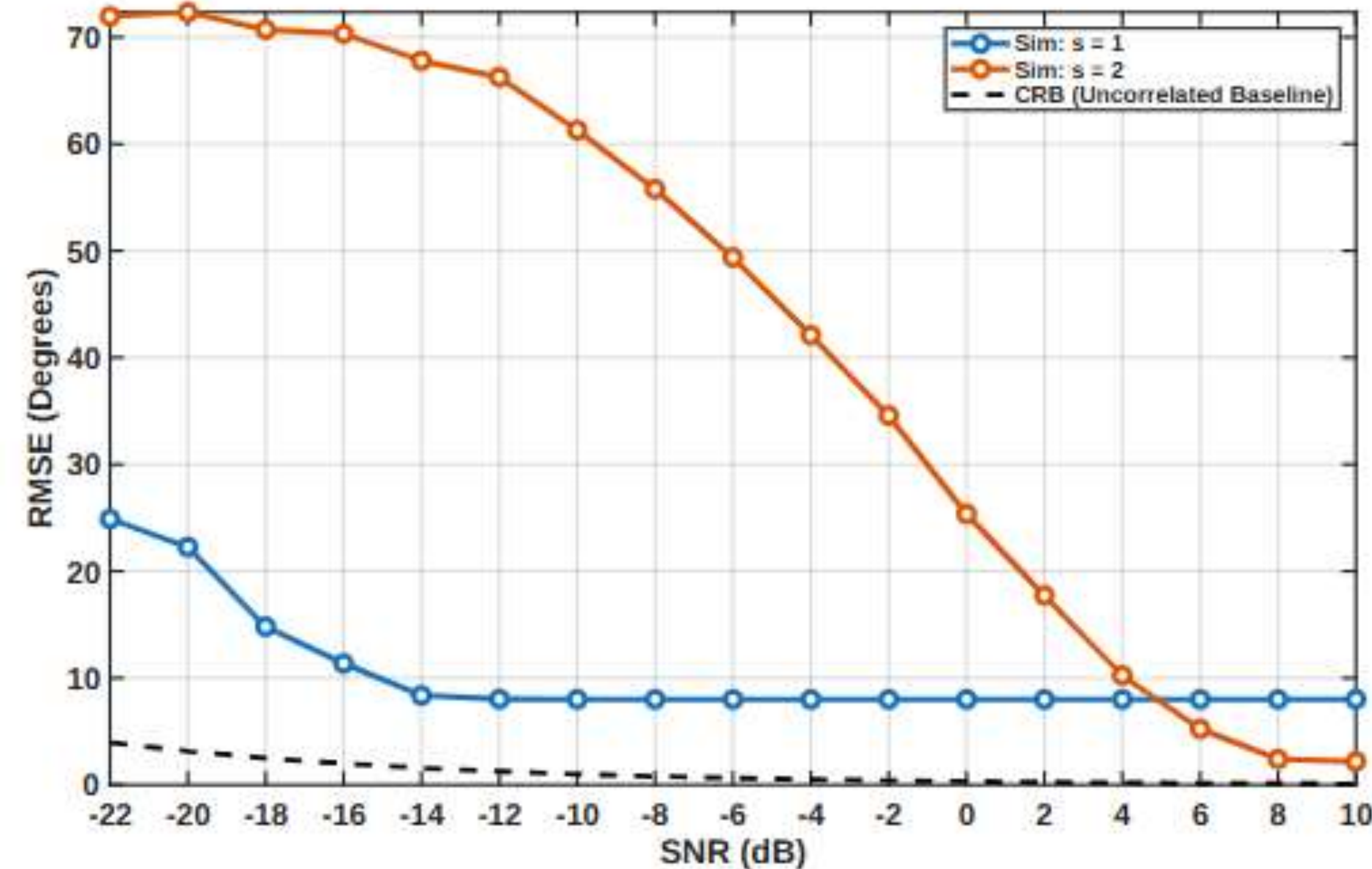


Fig. 6: RMSE comparison between standard UCA-RB-MUSIC ($s = 1$) and the proposed mode-space smoothed variant ($s = 2$) against theoretical CRB.

- Fig. 7 quantifies the operational limits of the proposed algorithm, demonstrating an observed tolerance of up to +4 dB.

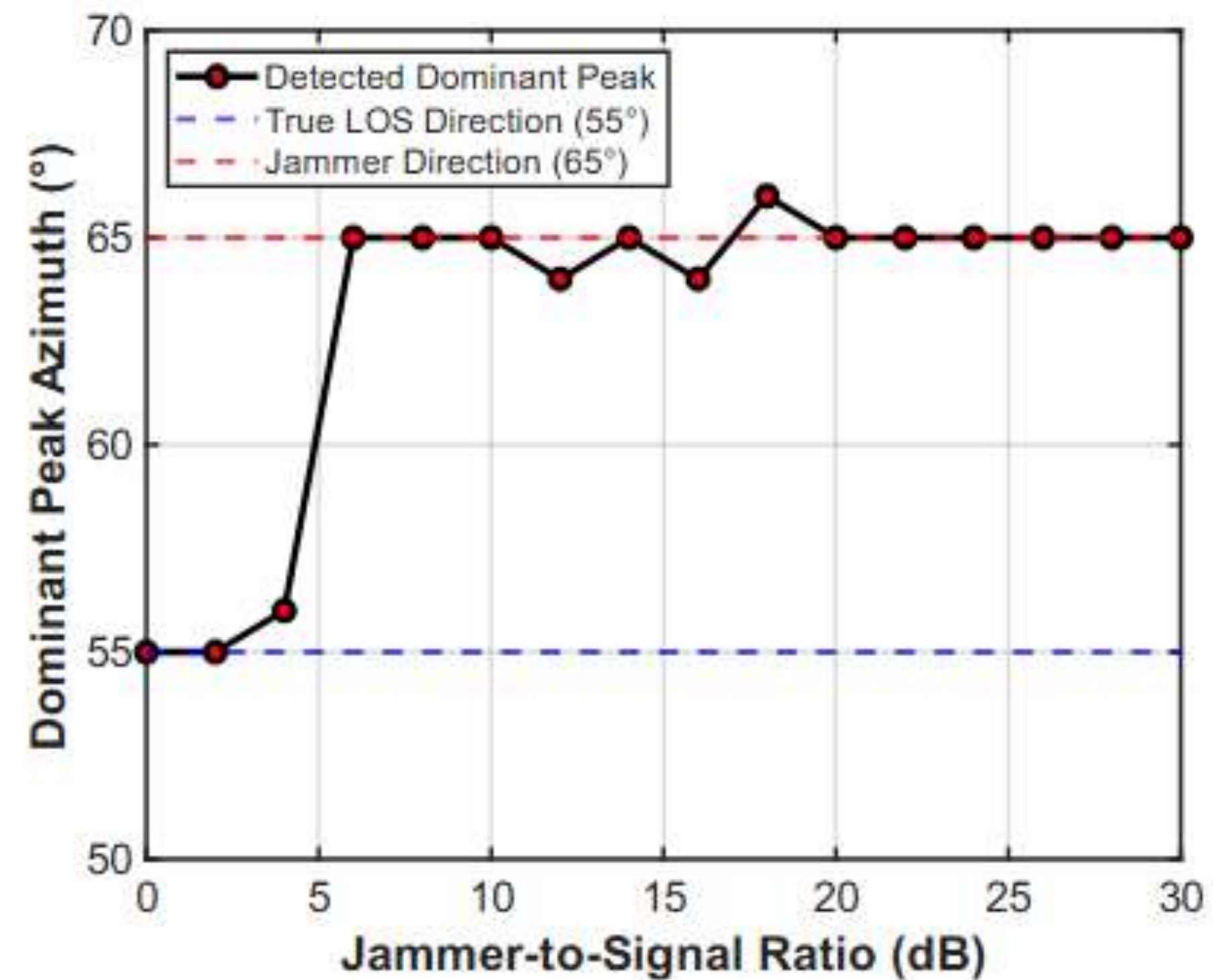


Fig. 7: Dominant detected azimuth as a function of jammer-to-signal ratio(JSR).

Scenario	Channel Condition	Delay [S1, S2, S3, S4]	Peak (φ, θ)
Open Terrain Combat	Specular-dominant	Short: [0, 1, 2, 3] samples	(55, 90)
Open Terrain Combat	Specular-dominant	Medium: [0, 10, 20, 30] samples	(55, 90)
Open Terrain Combat	Specular-dominant	Long: [0, 100, 200, 300] samples	(65, 90)
Urban Combat	Fully scattered	Short	(55, 90)
Urban Combat	Fully scattered	Medium	(65, 90)
Urban Combat	Fully scattered	Long	(65, 90)

Table II: First Peak Estimates across Open Terrain Combat Zone and Urban Combat Zone Scenarios

- Table II summarizes localization across both channel conditions. Longer delays cause spatial decorrelation which in turn degrades the algorithm performance.

- Open Terrain: $h_i = \begin{cases} r_i e^{j\phi_i}, & i = 1, 2, \\ r_i \frac{X_i + jY_i}{\sqrt{2}}, & i = 3, 4, \end{cases}$ Urban: $h_i = r_i \frac{X_i + jY_i}{\sqrt{2}}, \quad X_i, Y_i \sim \mathcal{N}(0, 1).$

Appendix: Minimum Phase

Case	Channel Condition	Delay Spread	Model Order Selection Criterion	Estimated peaks (φ , θ)	Power (dB)
Unsmoothed	Specular Dominant	Short	Manual	(61, 57) (188, 49) (323, 89)	0 -3.87 -3.99
	Specular Dominant	Moderate	Manual	(62, 56) (322, 89) (187, 49)	0 -3.72 -4.04
	Fully Scattered	Short	Manual	(58, 61) (65, 52) (271, 39)	0 -4.68 -5.51
	Fully Scattered	Moderate	Manual	(59, 59) (188, 43) (321, 81)	0 -4.11 -4.43
Smoothed	Specular Dominant	Short	Our Method	(65, 54) (55, 29) (87, 37)	0 -4.8 -11.85
	Specular Dominant	Moderate	Our Method	(65, 54) (56, 62) (87, 37)	0 -6.66 -8.91
	Fully Scattered	Short	Our Method	(65, 54) (55, 63) (86, 37)	0 -4.98 -9.51
	Fully Scattered	Moderate	Our Method	(65, 54) (55, 63) (86, 37)	0 -12.69 -14.46

Ground Truth Signal Scenario:

Source	Φ (Az)	Θ (El)	Power
S1 (LOS)	65	54	0 dB
S2	55	63	-3 dB
S3	85	38	-6 dB
S4	100	22	-12 dB

($\theta \neq 90$)

Thank You!

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